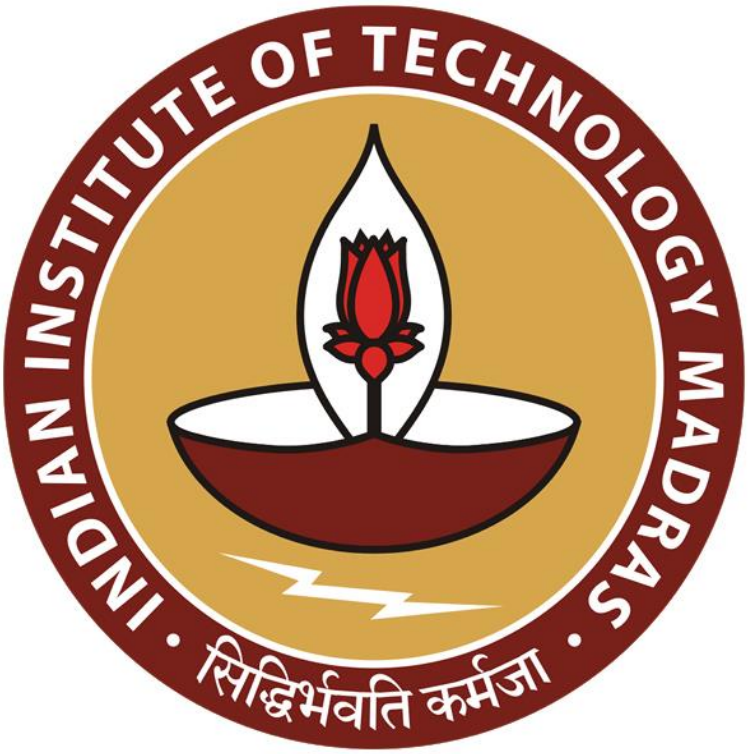
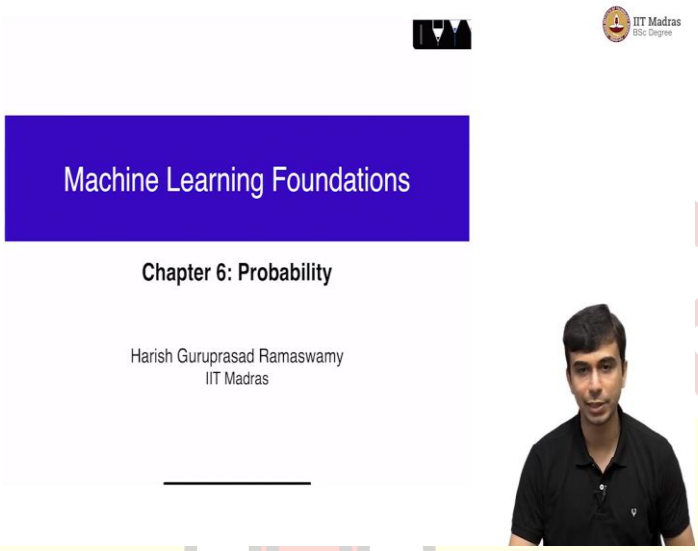


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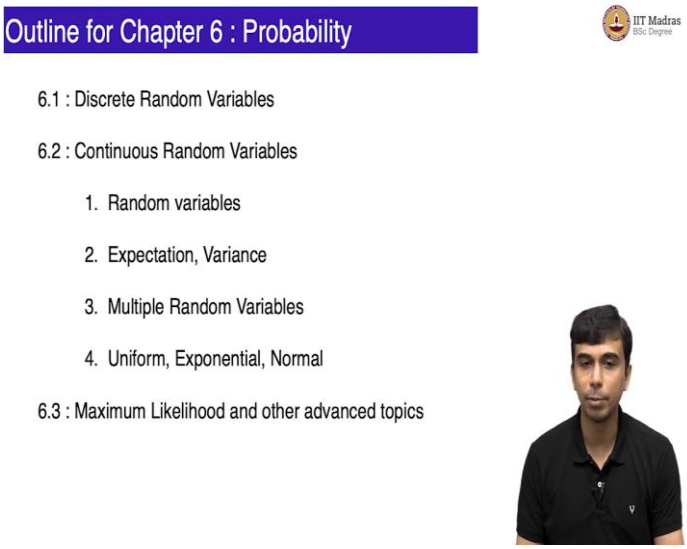
Professor Harish Guruprasad Ramaswamy
Department of Science and Engineering
Indian Institute of Technology, Madras
Continuous Random Variables

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Hello, everyone. Welcome to the second subsection of the probability chapter in which we will look at continuous random variables. We have seen some examples and concepts related to discrete events and discrete random variables, will, a lot of those ideas will extend to continuous random variables as well. But there are some extra subtleties and important things which we will see here.

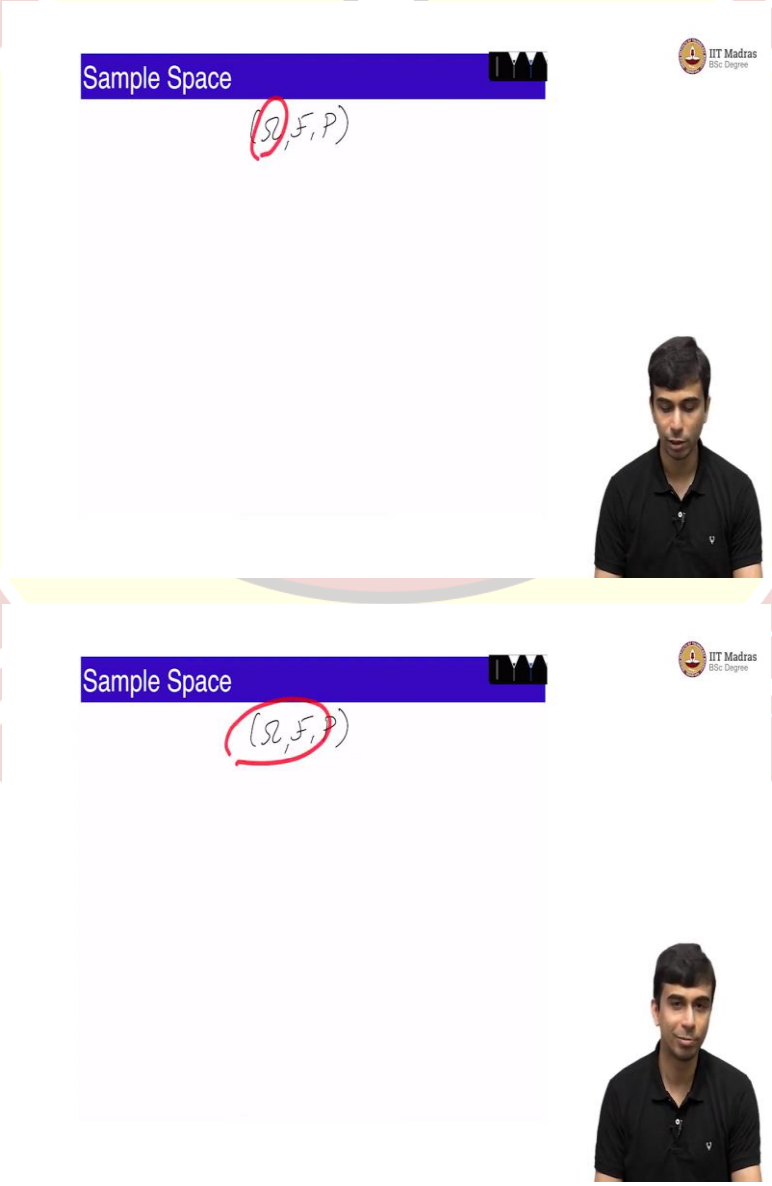
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Here is a brief overview of what we will cover in the continuous random variables. First, we will just define continuous random variables, and the basic ideas which, some of which you would have seen already in the discrete random variables. But, how do you extend those ideas to continuous random variables?

And some of these are very similar to the discrete random variable case, you can see that there is expectation and variance, multiple random variables, and so on. These were all ideas that were available in discrete random variables as well. So, I would highly recommend that each concept that you see in continuous random variables, compare and contrast with discrete random variables, how are they similar, how are they different, and so on.

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So, the first thing is sample space, we have already seen that an experiment is made of a probability, an experiment in probability is made of three things, a sample space omega, and collection of events F, and the probability measure P. In the discrete random variables case, we have almost always been using omega's finite set, or at most a discrete in finite set. So that is what we have always worked with.

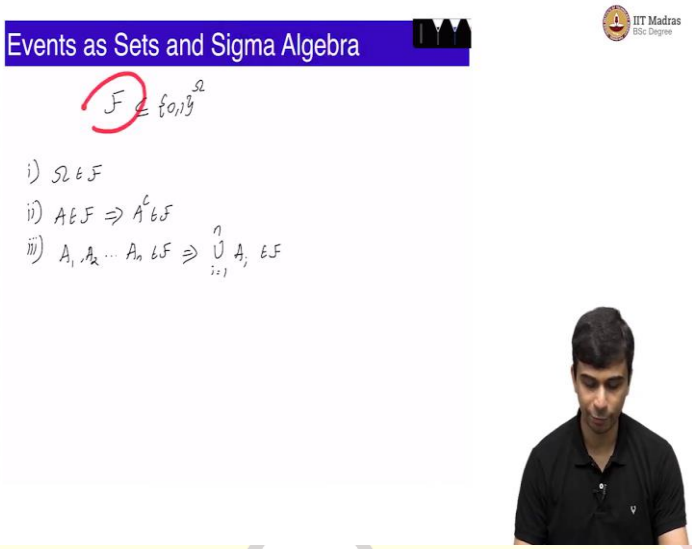
But in general, there is nothing restricting an experiment to how only a finite discrete amount of outcomes. For example, let us say, you pick a random person in the world and measure their height. The output of this experiment is not discrete. So, it can be mean, the height of a person can be 6.1311 meters, and so on. So, this is an example of an experiment where the sample space is continuous.

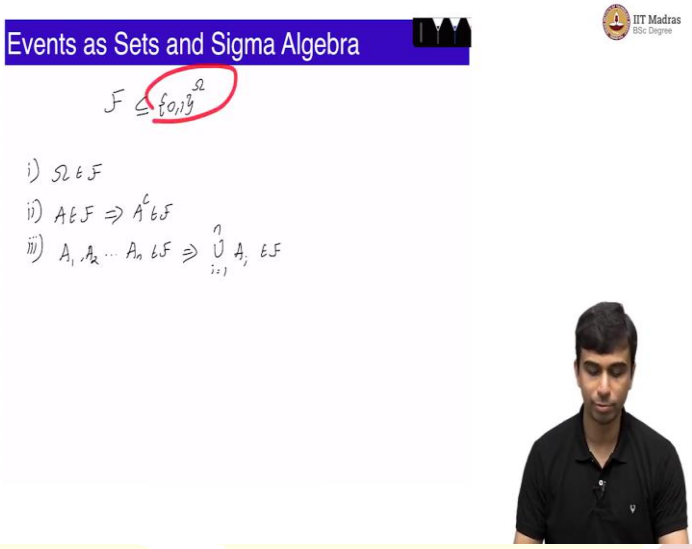
Of course, you could argue that everything is discrete at a quantum level, and so on. But let us ignore those technical issues. And imagine a case where you can have really continuous set of outcomes. So that is the, that is your omega can have infinitely uncountable outcomes, also, that is an example.

That is how, that is the first requirement for studying continuous random variables, if omega happens to be discrete or finite, then there is no such thing, while all variables associated with such an experiment are going to have to be discrete. But even with the experiments, where omega the sample space is continuous and uncountable, you can have discrete random variables, we will see.

But if you must have omega to be uncountable, and continuous for you to even consider, mean for you to even have continuous random variables. So, in this particular module, we will always have our sample space omega, to be a continuous or infinitely uncountable set.

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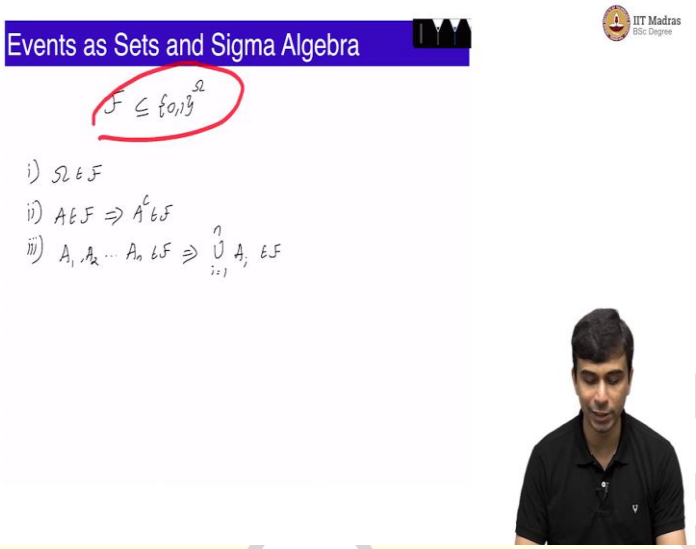


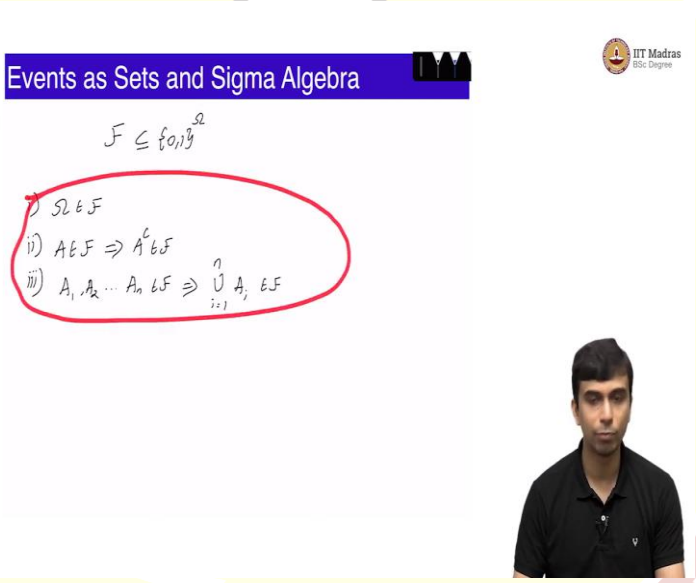


So, that brings us to another an important complication, which is our F. Previously, this is what we had used when our omega sample space omega was either finite or just discrete. We just used our set of events, as simply the set of all events, which is 0 , one power omega, that is what we have been using. But that is no longer possible.

If your omega is for example, let us say all reals or let us say you are measuring someone's height, potentially, any real number, any positive real number is a possible output of someone's height. So, you are omega is essentially the entire real line.

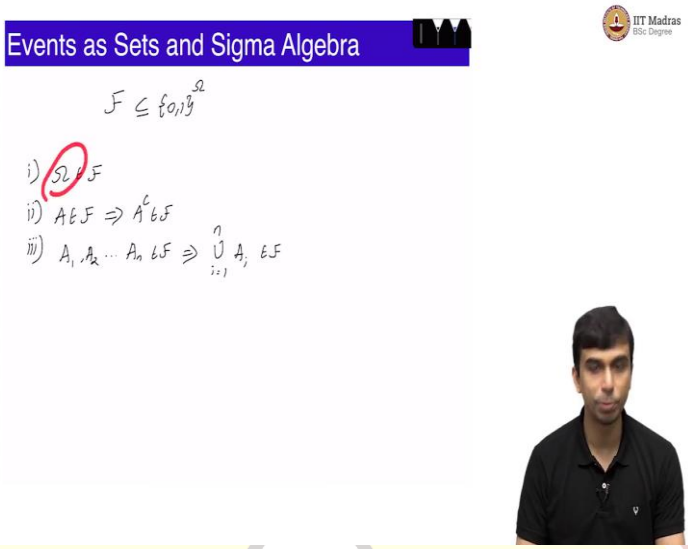
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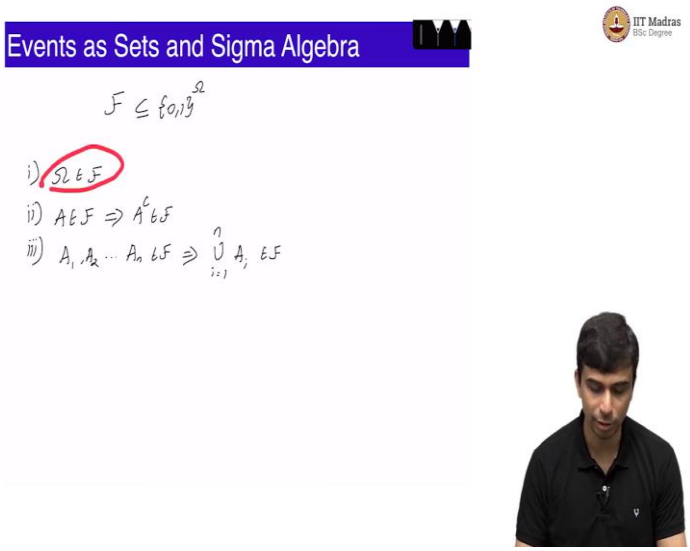
If that is the case, you cannot have 0 , 1 power omega, all subsets of the real line is not a valid, there are issues with choosing that to be the set of all events. So technically, what people do is you choose a subset of that. So that is why, your collection of events F is a subset of 0 , 1 power omega. So, it must satisfy certain properties. What are the properties? Well, their properties are pretty simple.

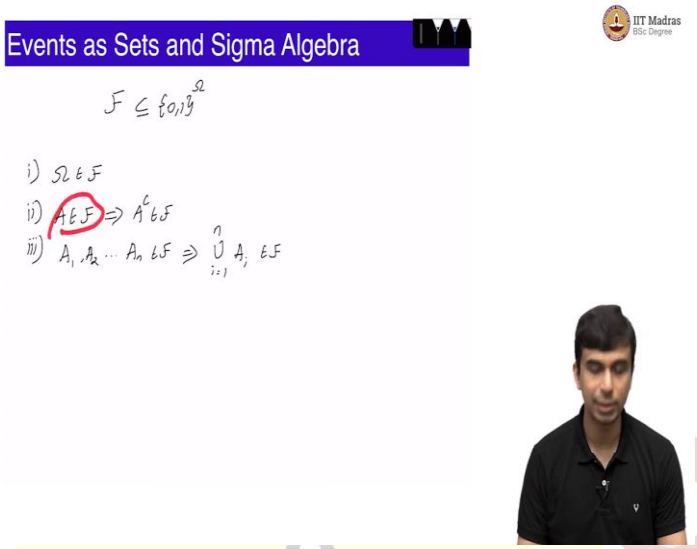
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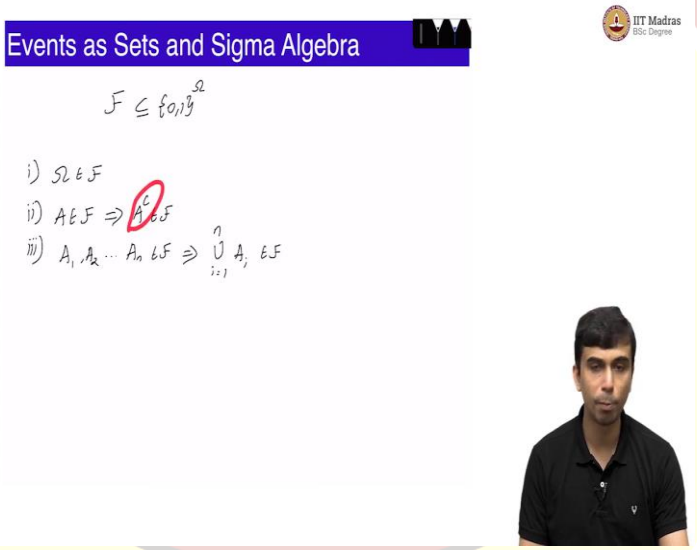


First is that omega must be in that. So, this is, all of these properties are going to hold. Your Omega was discrete and F simply 0 , 1 power omega, that is the power set of omega. But we do not, we cannot do that for continuous omega. What we will do is, we will actually work with a sub set of all, subsets of the power set of omega. And, what all are valid collection of events? Well, it must satisfy these properties.

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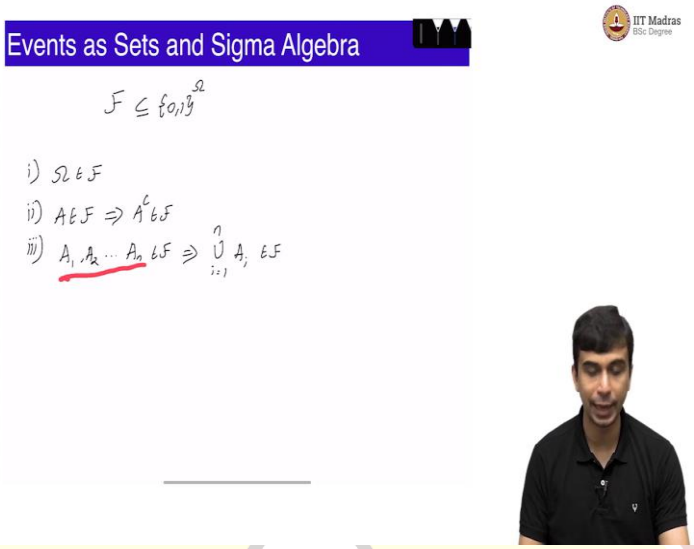


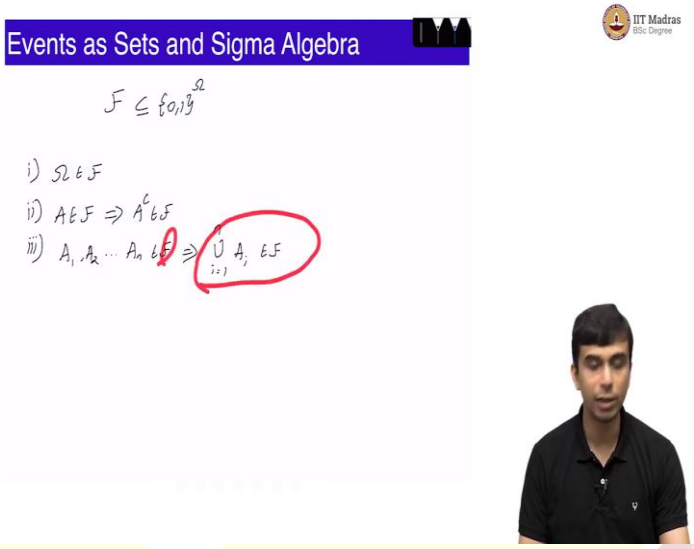


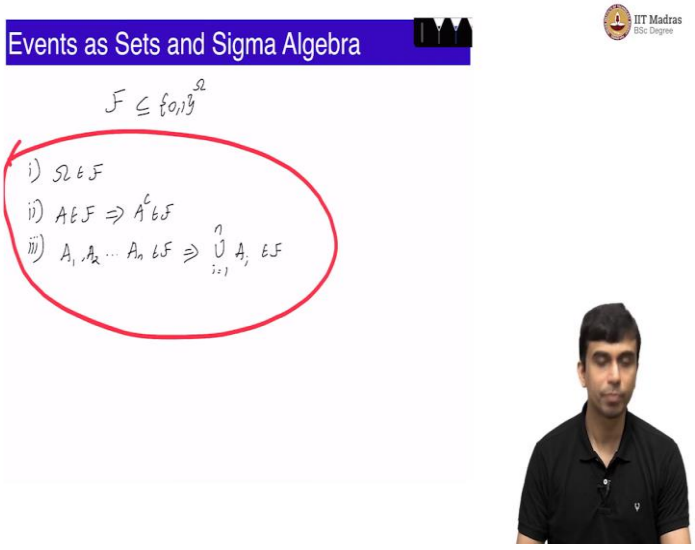


The first is that your sample space must belong to the collection of events. And then, if there is some set A which belongs to the collection of events, then A complement must also belong to a collection of events.

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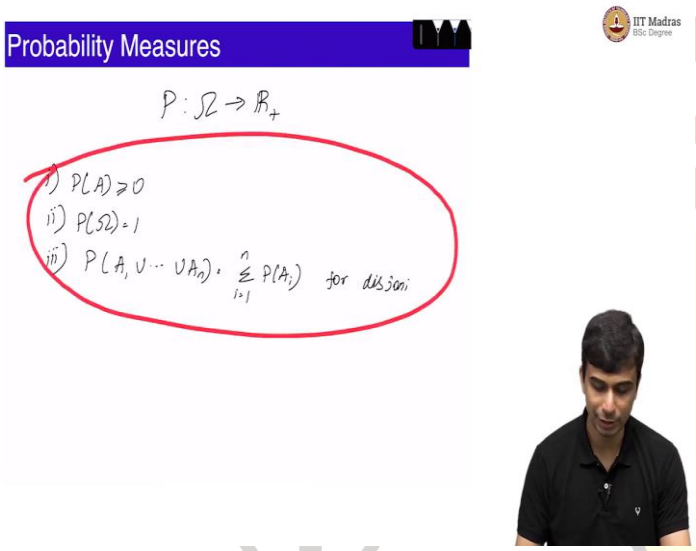


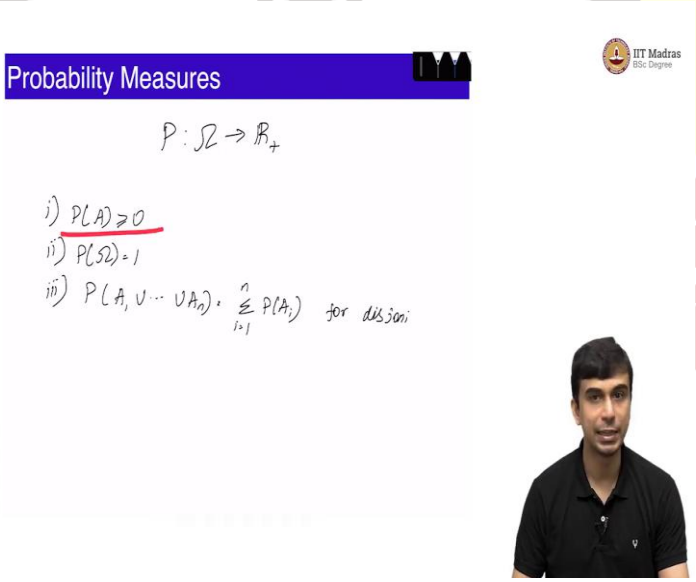


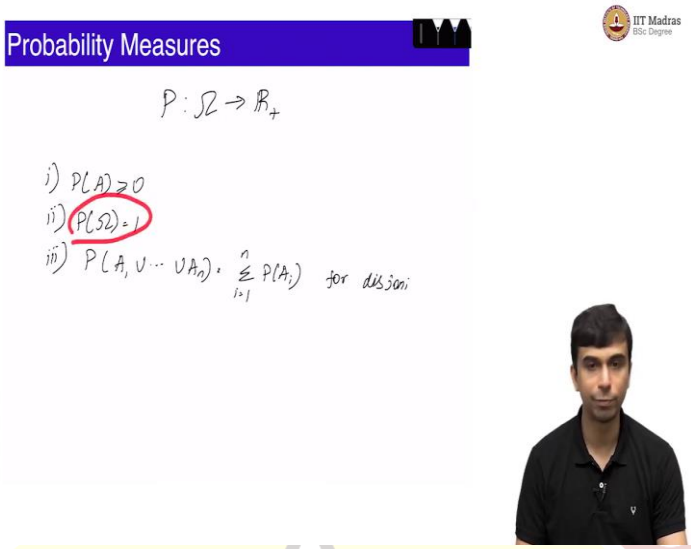


And similarly, you have A1 A2, An belongs to the collection of events. Then the union of that must also belong to the collection of events. So, that is, these are, these are mostly not, these are technical considerations. And mostly you do not need to worry about it at this stage. Because it is your first probability course, you do not need to worry about this. But you must understand that only events, collection of events will satisfy these axioms are allowed, these are called as sigma algebra axioms.

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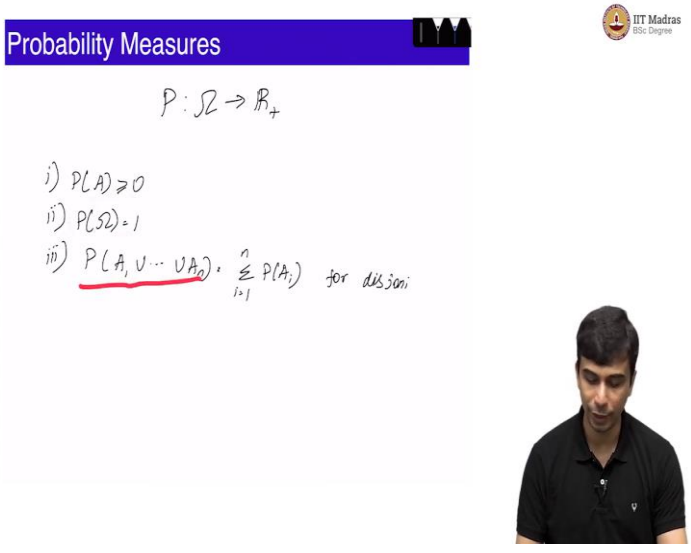


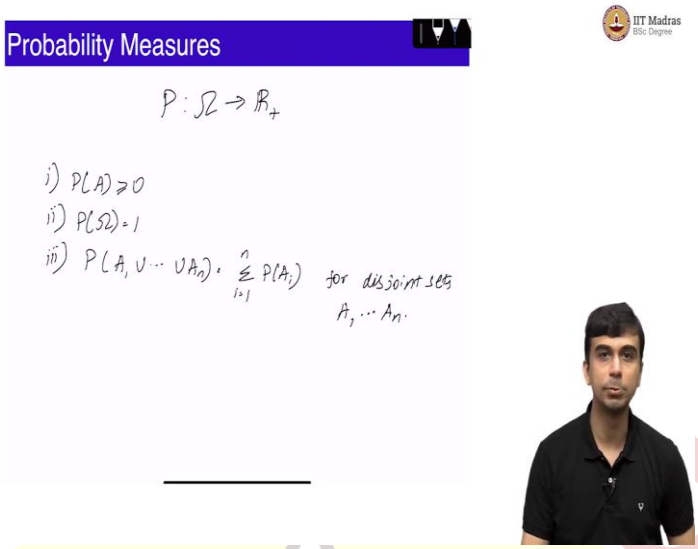




Then we have the probability measure, the probability measure is once again the same as before is a mapping from the sample space omega to R + all, it is the same as before. And it has the same axioms as before, these axioms are not new, it is exactly the same as before that is, you have probability of any set A should be greater than equal to 0, and probability of omega should be equal to 1.

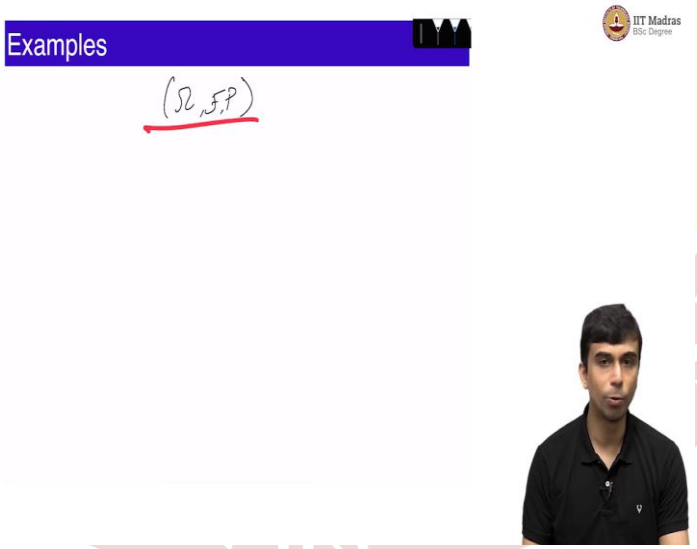
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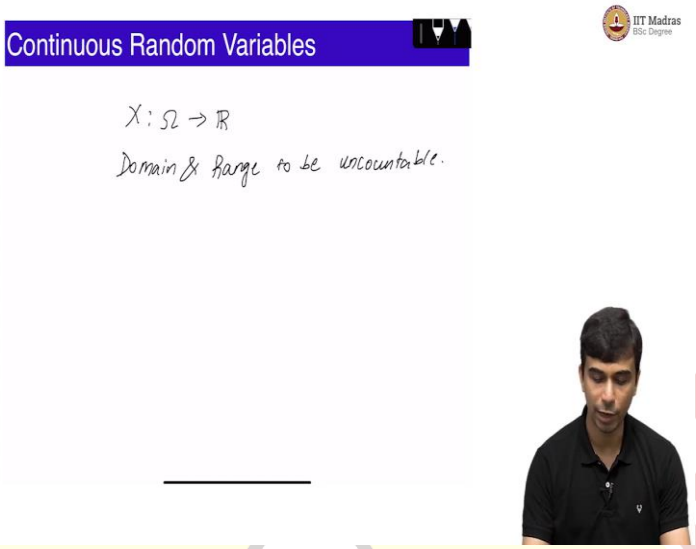
And if you have A1 to An are disjoint sets, then the probability of the union of disjoint sets is simply equal to the sum of the probabilities. So, these are the same axioms that we had for discrete random variables. But still, I am going to emphasize them just for the sake of completeness.

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Once again, we have our omega F P as the tuple, which represents the probability space or the experiment. So, all random variables are always in the context of an experiment and experiment is defined by omega F and P.

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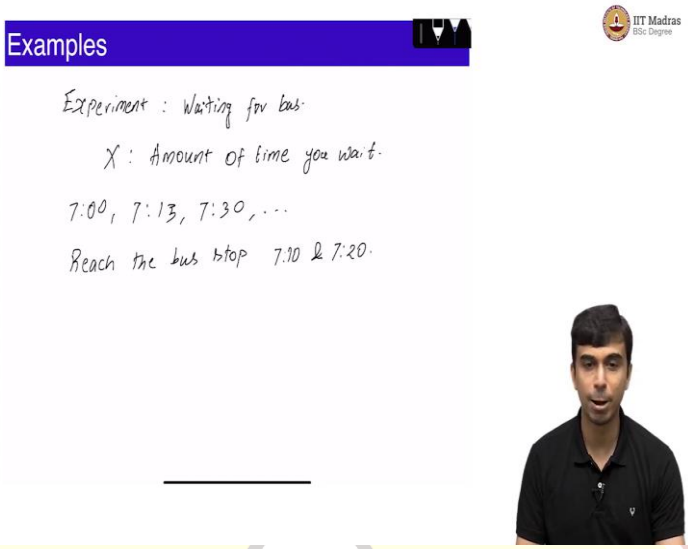
So, I am not really going to go into great detail here. Now we will move on to the main player here, which is continuous random variable. A continuous random variable, random variable is defined in exactly the same way that is, is a function, which maps your sample space omega to the reals R. That is exactly the same, but the difference is that your omega is going to be infinitely uncountable, and your R is also which means that your domain and range, R domain and codomain are both in finite.

Unlike the previous case, where the discrete random variables that are domain and range and your codomain were finite, and domain and range are also finite. But here we have that X as omega, as an (())(7:38) omega to R. We have both domain and range to be uncountably, infinite. So, if you do not understand what uncountably infinite means, just take it to mean the set of all real numbers.

That is an example of an uncountably infinite. The set of integers is a countable infinity; the set of real numbers is an uncountable infinity. That is the definition of random variables the same just that the what is the difference between continuous and a discrete random variable, a discrete random variable has the range of a discrete random variable is countable.

But the range of a continuous random variable is not. There are some extra properties also. But that is one of the main differences. So that is a random variable.

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Let us see some simple examples. For example, let us take the experiment that we had, let us say the experiment is that you are going to catch a bus. So, that is the experiment and the time at which you go to the bus stop is the random is the, is the, is the source of randomness. Anytime you have a, an experiment, you should always ask what is the source of randomness. For example, if you toss the coin, what is the source of the randomness?

The source of the randomness is that, is simply chaos. So, you toss the coin, you do not really know what is going to happen. Similarly, this case for, let us say, throwing the dice. So, any experiment, what is the source of randomness? The source of randomness can mean, can be mostly, let us say if it is physics, it is going to be chaos or something like that. Or it can just be things that are beyond your control.

Or things that you do not know how to model. For example, let us say you are modeling stock prices. Whether that is your experiment, you are experiment, you can ask the question, your you can experiment is just simply let us say, observing the stocks stock prices over the day. Even though technically speaking, there is no randomness if everyone, it is simply a deterministic function.

As far as you are concerned, it is randomly because it is the modeling the stock prices is a deterministic function is impossible. So that is the reason why you model it as a random variable. So, let us say in this example, your randomness is the time at which you reach the bus stop. So that is the randomness. And there is let us say there is a bus every 15 minutes. So, you can have the random variable, X.

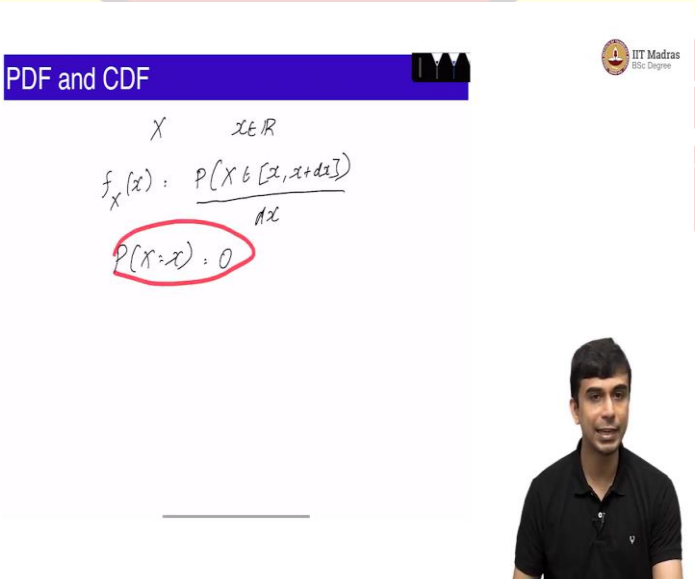
I am not going to write, (())(10:18) what the experiment is. Omega, the experiment is, and the experiment is simply waiting for the, waiting for bus. Let us call this waiting for bus. That is the experiment. And the random variable, now x is the amount of time that you wait for the bus, let us say. Of course, we need more details for this. Let us say that there is a, there is a bus every 15 minutes. So, there is a bus at 7:00 o'clock, 7:15, 7:20, and so on.

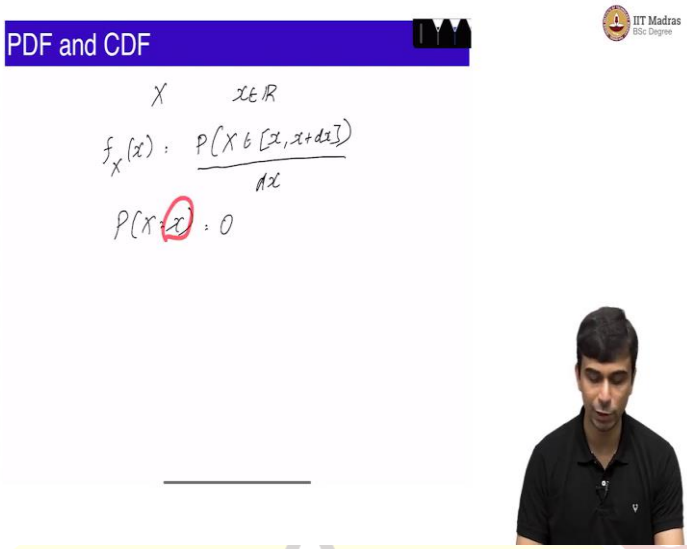
And, let us say you reach the bus stop anytime at random between 7:10 and 7:30, so this is a pretty standard experiment. So, you have bus coming at, let us say 7:00 o'clock, 7:15, 7:30, and so on. And you reach the bus stop anytime at randomly between 7:00, let us say 7:10, and 7:20. This is the experiment.

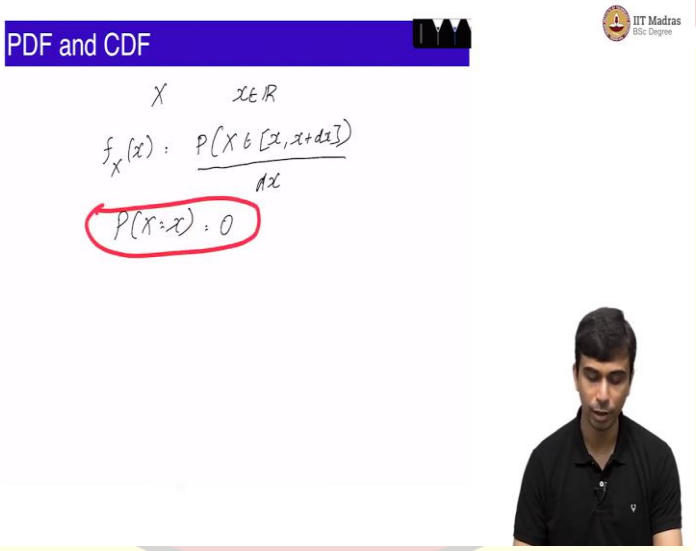
And the experiment is simply a manifestation of the real world and the random variable. In this experiment, let us say you come to the bus stop, and you wait and you catch the bus and you go. So, one experiment is this. The random variable of interest, let us say in this case, is the amount of time you wait for the bus.

So, what would this take, it would take the results of this experiment, omega would be after this experiment is done. Your X is a random variable which takes in the result of the experiment, which would essentially take into account when you arrived, when you caught the bus and so on and could give you how much time you waited for a bus. This is an example of a random variable. So, that is pretty simple example. And that is sufficient.

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So, the same way that you have the properties and attributes of a discrete random variable, you can have properties and attributes of continuous random variables also. And the clearest, we will see several analogs. The first analogue is, what is called the density function. So let us say you have a random variable X, the density function is $f_X(x)$.What is the meaning of this of let us say you have some x in reals.

What is f_X (x)? f_X (x) is essentially, the probability of x being very close to x, let us say x to x + dx, where dx is some arbitrary small-time interval. In the case of the bus experiment, dx is arbitrary small-time interval, divided by dx. That is why f_X of x is called a density function, it is the probability density function.

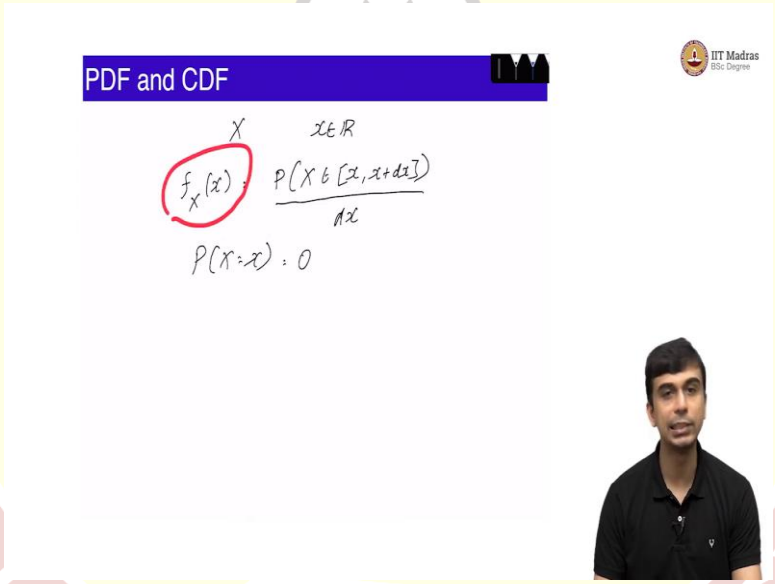
And this is really important because you cannot use a probability mass function. Why is that? Because let us say probability of X is equal to x is equal to 0 for any x. Why is that, because

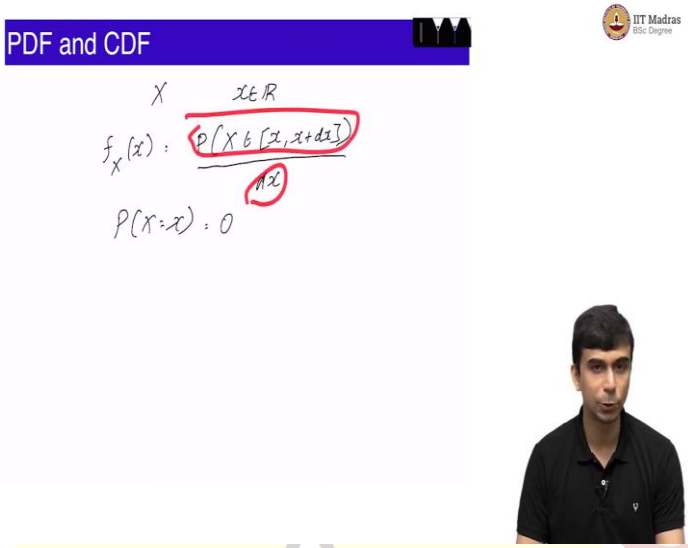
for any given time, instant probability of you arriving exactly at the time instant is 0. Let us say the bus experiment, someone can ask the question, what is the probability that you exactly wait for 4 minutes and 29 seconds?

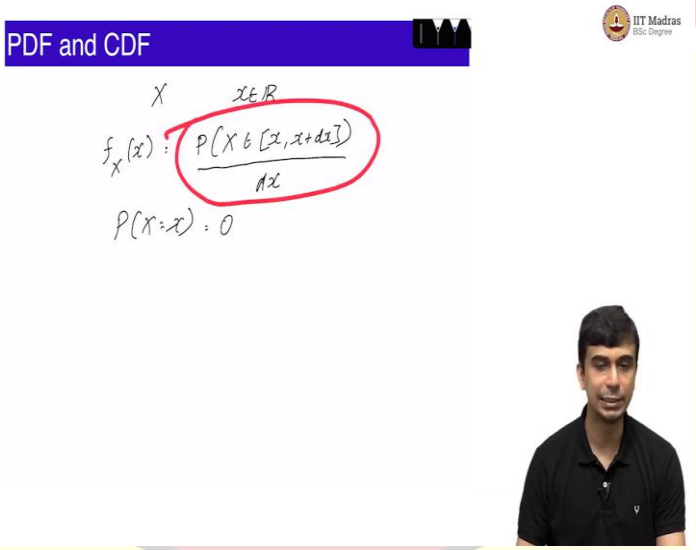
What is the probability wait for exactly that? Well, the answer is 0, because there are so many options. Why would you exactly wait for 4 minute 29 seconds, the probability of that event is 0? But someone can ask the question, what is the probability that you wait between 4 minute 29 seconds and 4 minutes 35 seconds? You can ask that question.

And that is, what might be small but still not be 0. So that is the reason why you have probability of any, if you have any continuous random variable, the probability of the continuous random variable taking an exact value x is always going to be 0.

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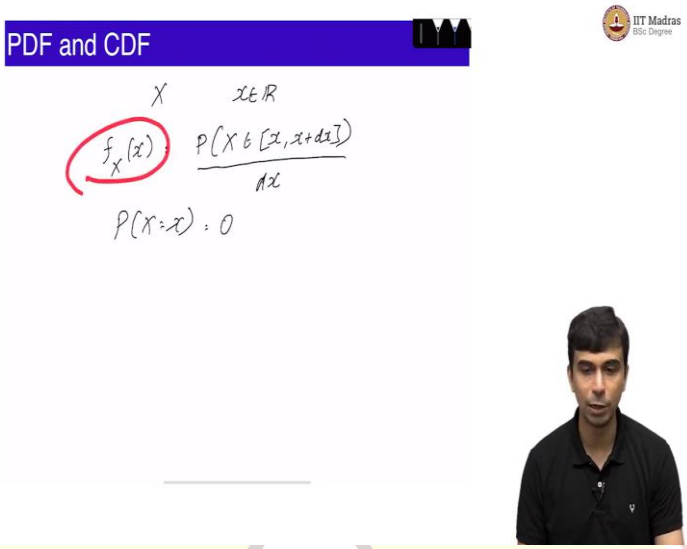


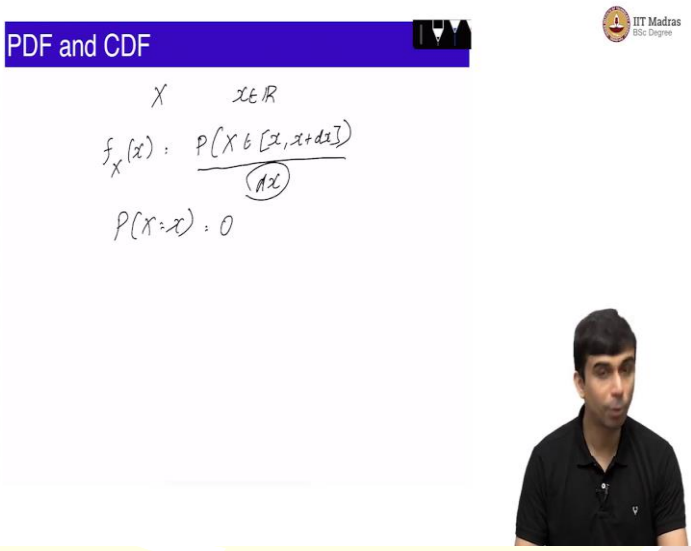




But despite that, you can have density around that. That is why, you have, you are dividing that by dx, so even though this numerator is going to be 0, the diameter is also going to be 0. So that is the idea behind a density function. So, you will know probabilities do not have units' density functions, in a way you could, can say they have units because they are dividing by x.

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For example, in the bus waiting experiment, when x is the amount of time that you wait for the bus, the unit of the density is going to be, let us say, per hour or per minute. So based on what you want to use can use per hour or per minute for, if x was the height of the person that you are measuring, will say in height measuring experiment, X represented the height of the person that you are measuring, the density would be, let us say, per meter, the unit of that would be dense because you are dividing by dx, probability does not have any unit.

But when you divide by dx, the unit is automatically created. That is okay. So that is the reason why you cannot have probability mass functions, you need to have densities and that is the reason why you will have for continuous random variables, probability mass functions do not make sense you only have density functions.

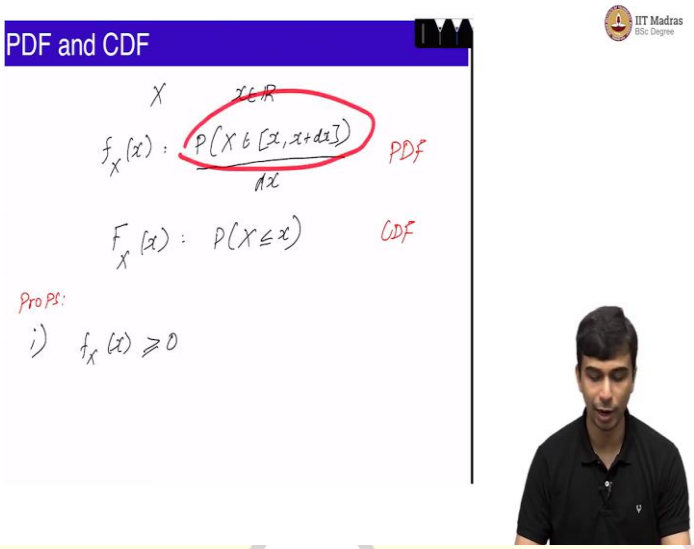
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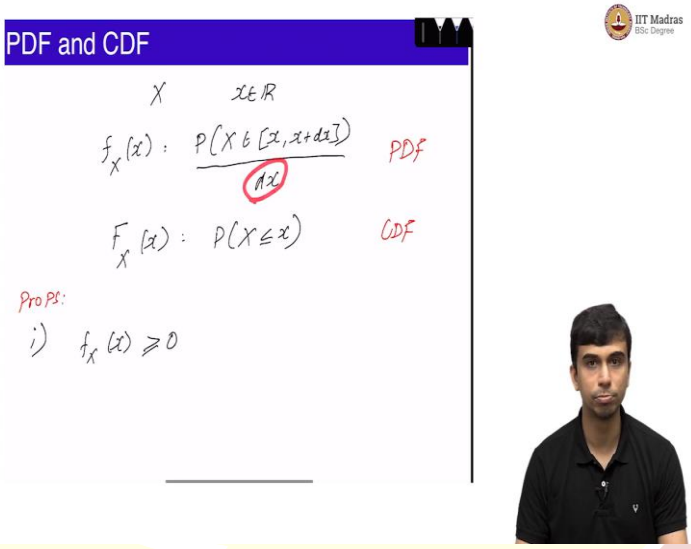


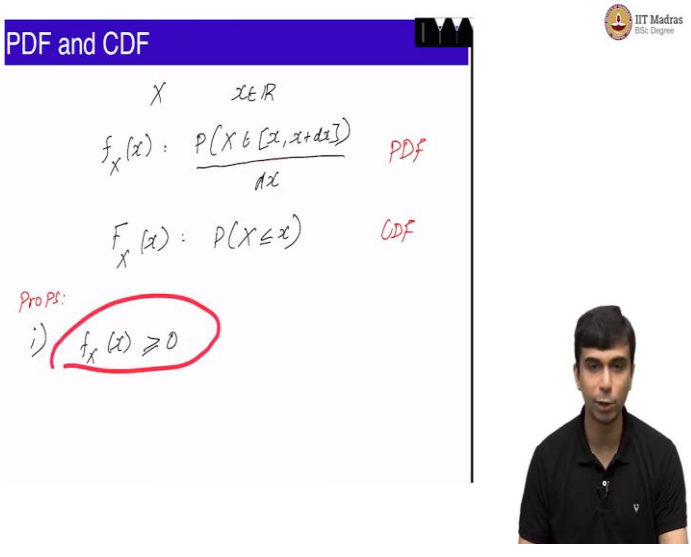
So, this is called the probability density function or the PDF. So typically denoted by small f subscript X, and you will see that in almost all the definitions for which there is a discrete random variable analogue, essentially, probability mass functions and broadly density functions behave similarly. So, you will, in a lot of places where you have in discrete random variables, where you have some over x FX of x, you will see integral FX of x dx.

So, the sum will be replaced by an integral, but other than that, you will see very similar behavior throughout. So, but however, you should understand the difference between a density function and a probability mass function. But the cumulative distribution function is still a completely meaningful object here. What is that? Well, it is exactly the same as what we had for the discrete random variable.

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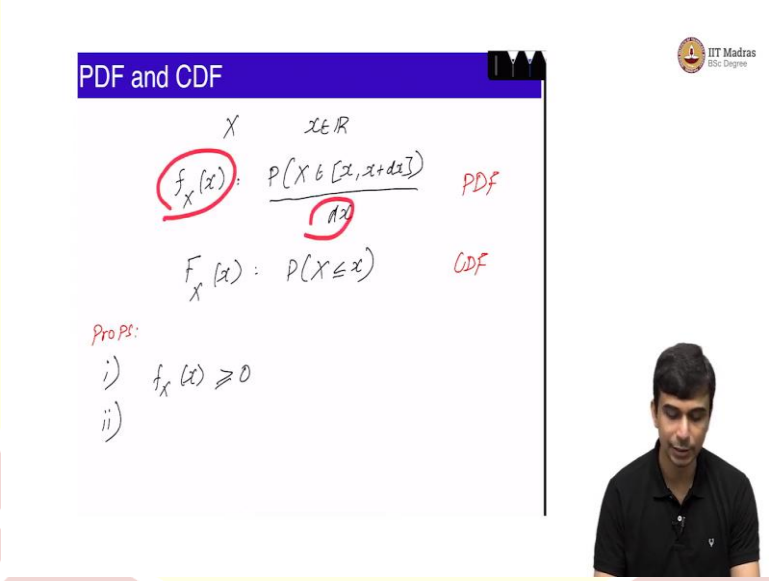


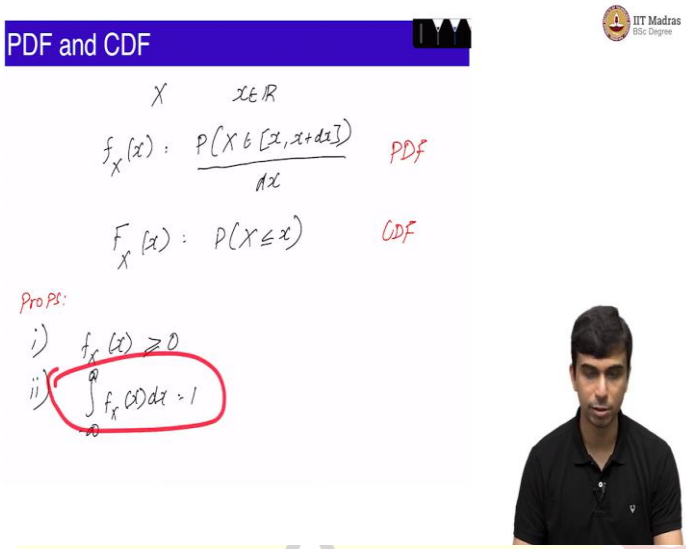
What is Fx of x? That is simply equal to probability of X less than or equal to x. This is the probability density function. And this is the cumulative distribution function. So let us see what are the properties of, let us give some important properties of the PDF and CDF. First property is that the density function is greater than or equal to 0.

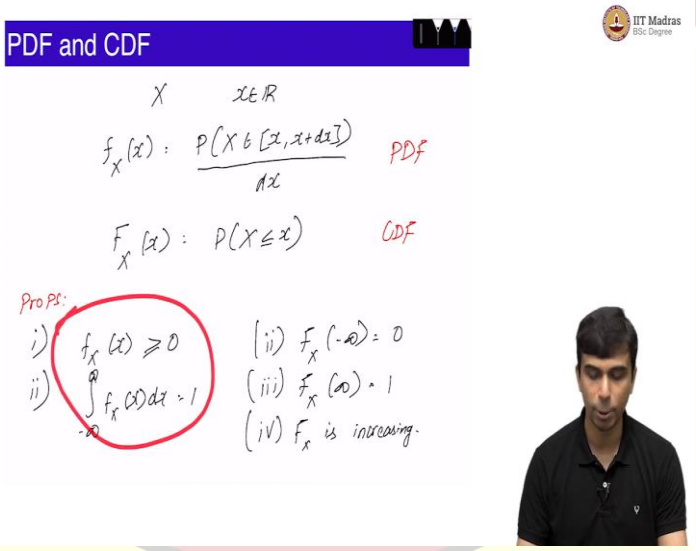
This is obvious because probability is always going to be greater than equal to 0, and dx is also positive. So, probability of X in x belonging to X + dx by dx is going to be positive. So, fX of x is greater than or equal to 0, but not however I am not saying fX of x is less than or equal to 1. Why is that?

That is because of the dx here, even though this number can be, this number can never exceed, the numerator can never exceed 1 because probabilities are going to be between 0 and 11, the denominator here, which means that the density can exceed 1. So that is the first property.

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The second property is similar to the analogue to the probability mass function summing to 1. The probability mass function sums to 1 if you sum over all the non-zero arguments of a probability mass function that sums to 1. And why is that, that is because the probability of the sample space, and the entire sample space is equal to 1.

And that is simple axioms of probability. By a similar logic, what you can have is fX of x times dx is simply probability of X belongs to x + dx, x, which means that to get the entire range, we just integrate. So that means the integral over - infinity to infinity, fX of x dx is equal to 1. This is the analogue of this properties, the analogue of the probability mass function summing to 1.

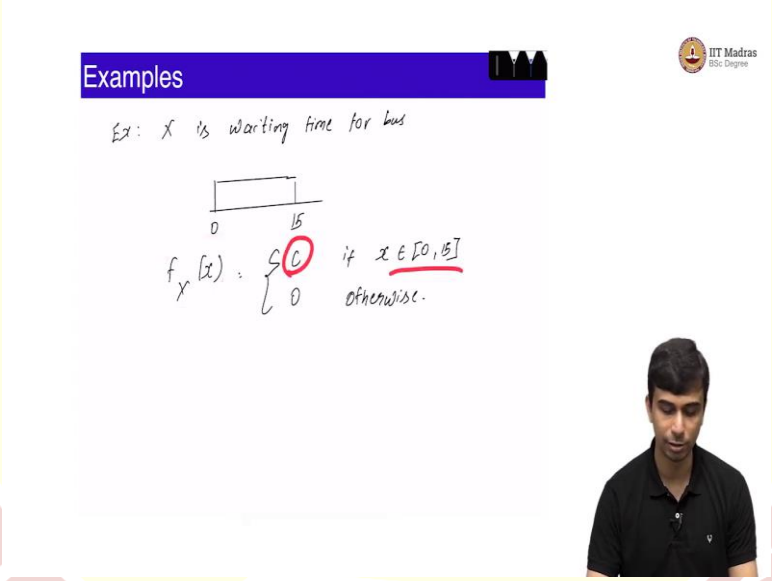
So, these two are the main properties of the probability density function. Are there any interesting properties of the CDF? Well, it is pretty much the same that of the discrete random

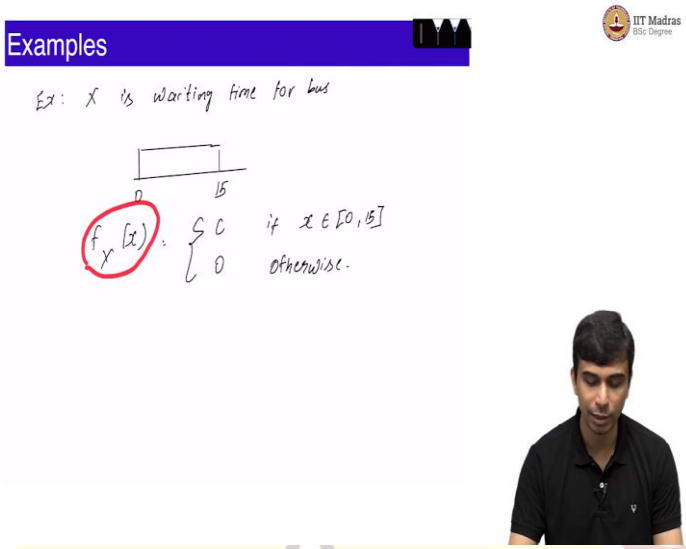
variable that is CDF would be FX of - infinity is equal to 0. And FX of + infinity is equal to 1, at the left end the CDF is 0, and the right end is one.

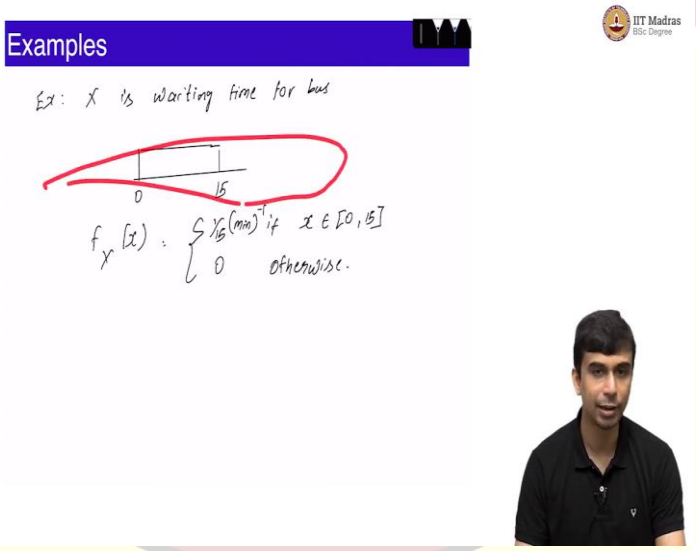
And there are other properties also, the same as the discrete random variable that is, CDF is monotonically increasing, it can never decrease. For example, if you have FX of 3.6, FX of 5 can definitely has to be greater than equal to 0.6. Why is that? Because probability of X less than or equal to 5 is definitely greater than or equal to probability of X less than or equal to 3, that is the reason why you have CDF FX is an increasing function.

I have to strictly say that it is non-decreasing, because it can stay the same, but I am going to say just increasing to make life easier for us. So, these are the main properties of the probability density function and the cumulative distribution function.

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Let us see some simple examples. X is amount of time, waiting time for bus, the smallest value is 0, and the largest value is clearly 15. So, let us make a simple, simple, simplification here, instead of you arriving uniformly between 7:10 and 7:30. Let us say you arrive between uniformly at 7:15, and 7:30.

So, if you are uniformly between 7:15 and 7:30, well the amount of time you wait, that is quite equally likely that you will wait for 0 minutes, which means that you will occur exactly at 7:30. Or you will wait for 15 minutes, which means that you occur exactly at 7: 15. So, this is a slightly simplified version, in which case, you will have the same number here.

Which means that fX of x is equal to some constant if x is between 0 and 15, and 0 otherwise. So, this is a simple example. Before let us take the more complicated example later on, where you are uniformly between 7:10 and 7:30. Let us consider a simple example where you are a

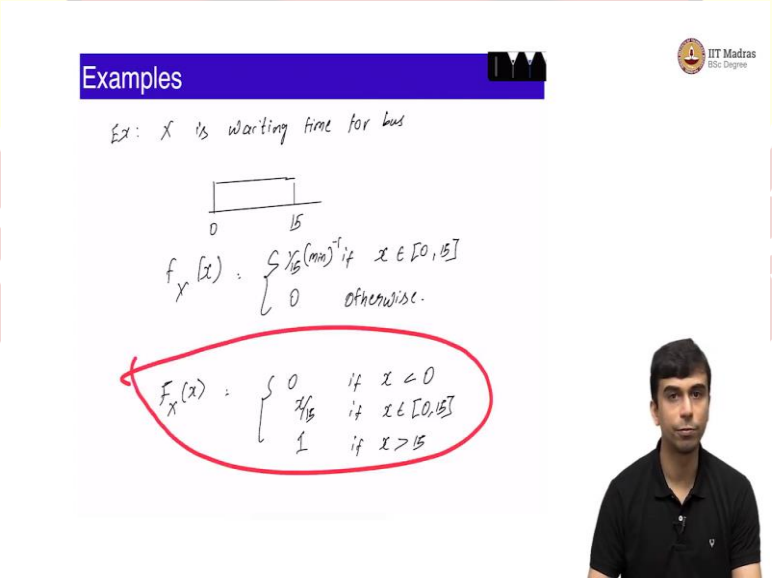
uniformly between 7:15 and 7:30. That is the thing. And how much time do you wait for the bus? Well, it is quite likely that you wait, any number between 0 and 15 minutes equally likely? So, that is a constant. But what is this constant?

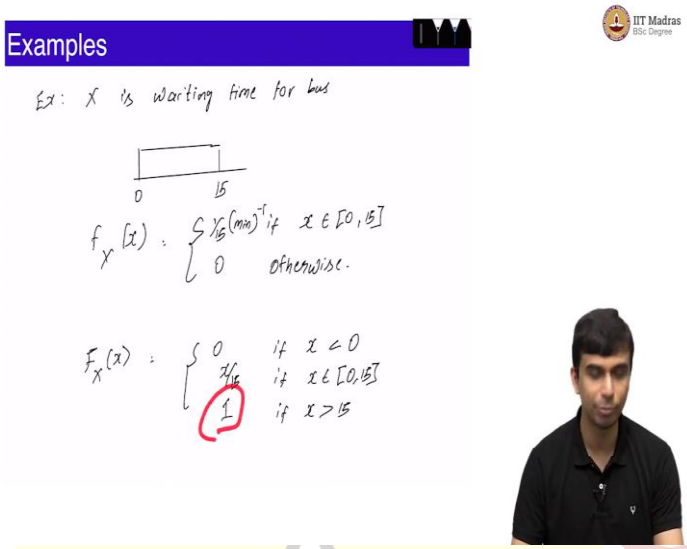
And second question, what is the unit in this case? Because I am using minutes here, the units here is fX of x ic C, per minute. So, you can take minute power - 1 is the unit for the density function here. But what is the value of C, here is where you can use the fact that when you integrate it over the entire real line, you are going to get, have to get 1, so which means that C is equal to 1 by 15. So that is the C 1 by 15 per minute.

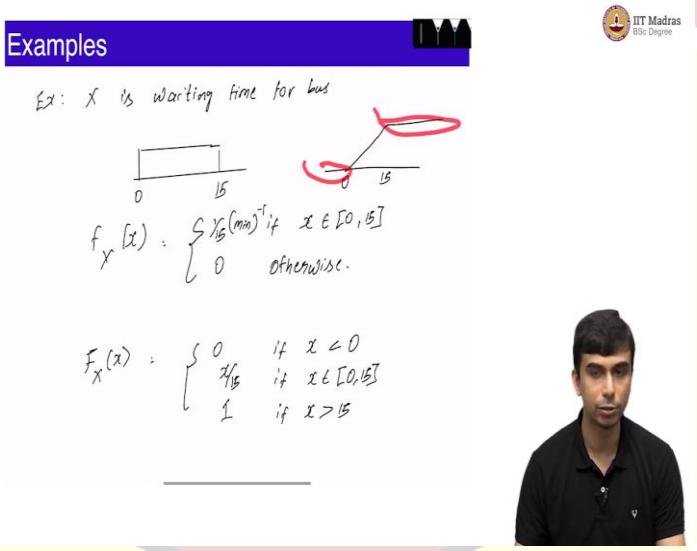
So, if instead if you had used, let us say, hour instead, and sort of minutes, the unit, you would see it is 4 per hour. That is 4 per hour for x between 0 to 2.5, sorry, 0 to 0.25. So, you will wait any time between 0 to 0.25 hours, the density is values 4. Why is 4? Because if you integrate this function from the entire, entire real line, it has to it has to be equal 1 which means that the area under this rectangle has to be equal to 1.

So, the height of this rectangle has to be simply equal to 1 by the width of the sample. So that is 1 by 15 per minute or 4 per hour. So, this is an example simple example of a density function. What is the cumulative distribution function, for this example?

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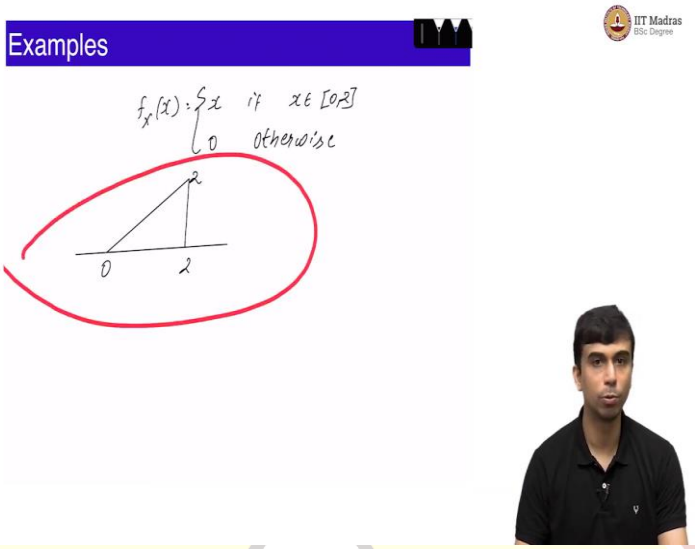


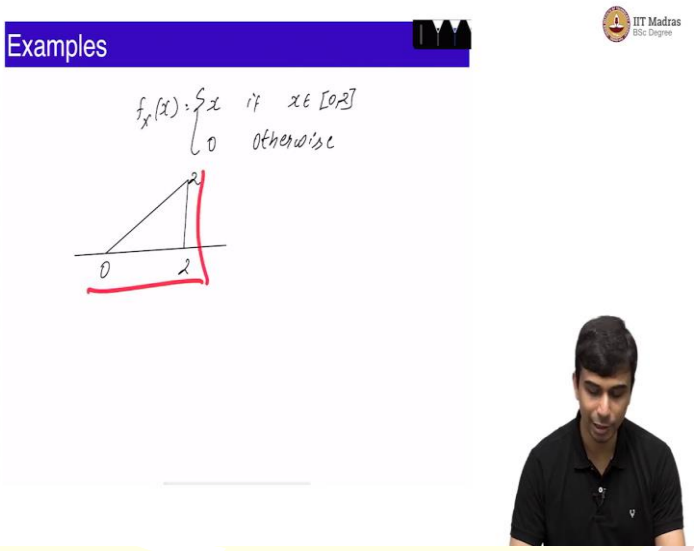


What is capital FX of x? Well, the answer for that is going to be, is going to be 0, if X is less than 0, so there is no way you can wait for a negative amount of time. So let FX of X is equal to 0. And then you have x by 15. If x belongs to 0 to 15, and 1, if x is greater than 15. This is an example of the CDF of this random variable where it rises linearly from 0 to the probability of x less than or equal to 0 is 0, probability of x less than equal to 15 is 1.

So beyond 15, you are definitely you are never going to wait for more than 15 minutes. So, that is why the probability of x less than or equal to 16 is the same as probability of x less than or equal to 17 is equal to and so on. If you plot at this, it would look like this. So, this is 0, this is 15. The value here, this value here would be 0. The value here would be 1, so this is the CDF of this random variable.

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Let us take one more example. Let us say you have your fX of x is equal to x, if x belongs to 0 2, and 0 otherwise. So, this is a density function. First of all, let us try to plot this 0 2. So, this is the density function. First of all, you should check whether this is a valid density function. How would you do that? Well, if you integrate it, you should get 1.

So, what is the integration of this function? Well, it is simply the area under the curve. And this is a nice triangle. So, what is this? So, it is half into base into height, which seems to say that half into 2 into 2 which suggests that this is not a density function. So, you should immediately see that this is a not a valid density function we have explained to instead.

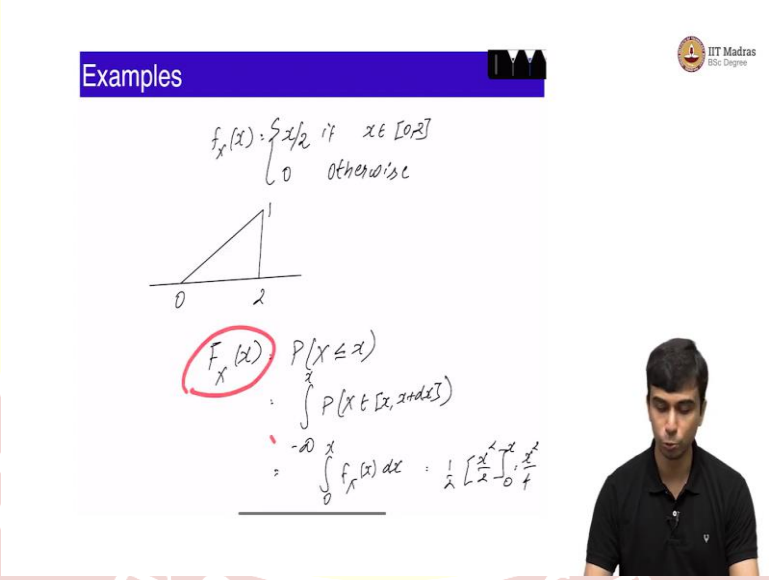
This is a valid, let us do function. Why is that because the area under the curve for this triangle is simply equal to 1. And what is the cumulative distribution function of this going to be, is

equal to probability of X less than or equal to x. And here is where you can use the calculus part of this, which is FX of x is probability of probability of X less than or equal to x, which is simply equal to integral - infinity to infinity, probability, - infinity to x, probability of X belongs to x to x + dx.

And this is exactly equal to integral - infinity to x, fX of x dx. And in this case, it will be x by so, - infinity can be replaced by 0 here because anything below 0 is, anyway, the density of x below 0 and 0, anyways you can just replace it by 0 x. And what is this value? This is simply equal to let us just consider x in the range of 0 to 2.

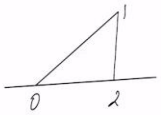
When x is less than 0, the CDF is 0 and x is greater than 2 is going to be equal to 1, the only interesting range for CDF is between 0 to 2 and within that range it is simply the integral over 0 to x, fX of x dx. And what is the integral half x squared by 2 within the limit, 0 to x. And this is exactly equal to x squared by 4.

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Examples

$$f_X(x) = \begin{cases} x/2 & \text{if } x \in [0,2] \\ 0 & \text{otherwise} \end{cases}$$

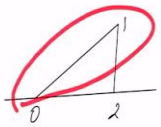


$$F_X(x) = P(X \le x)$$
$$= \int P(X \in [x, x+dx])$$
$$= \int_{-\infty}^{x} f_X(x)\,dx = \frac{1}{2}\left[\frac{x^2}{2}\right]_0^x = \frac{x^2}{4}$$



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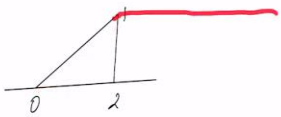


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You can check that any substitute FX of 2 you get 2 by 4 which is equal to 1. So, you have a CDF which is quadratic for a linear, linear density, linear density function. You have a CDF which is quadratic, it is 0 at 0 and at 1, at 2 it is 1 and after 2, it is constant at 1. And, so that is how the CDF looks like for this simple example.